

Precipitation behavior in a Nb-5W-2Mo-1Zr niobium alloy fabricated by electron beam selective melting

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ABSTRACT

Refractory niobium-based alloy Nb521 (Nb-5 W-2Mo-1Zr) was fabricated by electron beam selective melting (EBSM). The microstructure feature of as-built Nb521 alloy was systematically studied in combination with the unsteady thermal cycling inherent in the EBSM process. The content of precipitates gradually increased from the top region to the bottom region for EBSM-built Nb521 samples. Dot-like or rod-like precipitates with different morphologies were distributed inside grains or along grain boundaries in the as-built sample, whereas needle-like precipitates with a large aspect ratio existed in Nb521 alloy ingot. All precipitates in the EBSM-built Nb521 sample mainly contain (Nb, Zr)C and Nb₂C. With prolonging the thermal balance holding time, the elongated precipitates were partially broken, and became fine and uniform. It is expected that the results obtained in this study can lay a solid theoretical and technological basis for the fabrication of refractory Nb-based alloy by the EBSM process.

1. Introduction

The rapid development of aerospace puts forward an urgent need for capable materials that withstand extraordinary conditions in hot-end components [1]. Niobium alloys with superior high temperature strength [2–4], excellent corrosion resistance, and heat resistance [5–7] have become one of the promising materials for fabricating hot-end components. However, traditional metalworking processes, such as powder injection molding, vacuum-arc melting and welding, introduce numerous issues, for instance, long-term process, high tooling and mold requirements due to high melting point and poor oxidation resistance of niobium alloys [8,9]. Therefore, it is urgent to introduce a new method to fabricate niobium alloy rapidly in order to meet the growing needs for their components in the aviation industry.

Additive manufacturing (AM) processes are based on incremental layer-by-layer manufacturing, and offer advantages such as delivery time-saving and free designing, proposing an attractive route for forming aerospace intricate parts [5,6,10,11]. Electron beam selective melting (EBSM), as a kind of the powder-bed based metal AM processes, adopts a high-energy electron beam as the heat source, which has the advantages of high vacuum atmosphere [12], high energy absorption rate and low residual stress. A high vacuum degree is suitable for

forming reactive materials with high affinity for oxygen, such as titanium alloys [13,14], and the high energy absorption rate can enable the full melting of refractory materials with high melting point, such as tungsten [15,16], molybdenum [17], zirconium [18], niobium [19], etc. Low residual stress of components can improve the success rate and avoid waste samples caused by cracking. Meanwhile, the feature of rapid melting and rapid solidification inherent in the EBSM process will favor the formation of finer microstructure and precipitates than the traditional casting and forging processes. However, the morphologies and size of precipitates in EBSM-built samples will vary during layer-by-layer cycle heating and cooling, resulting in their heterogeneity of mechanical properties [20–22]. Therefore, to fabricate niobium alloys using EBSM is a challenging and promising project. To date, scant literature available was devoted to manufacturing niobium-based alloys by the AM process. Dense Nb components were fabricated by EBSM using bimodal powders [23], and the evolution of grain structure of as-built niobium alloys was studied. However, up to now, the precipitates are rarely mentioned in the literature, although they have a significant impact on mechanical properties of EBSM-built parts [24–26].

This study aimed to verify the feasibility of EBSM to produce the Nb–W–Mo–Zr system alloy, Nb521 (nominal composition Nb-5 W-2Mo-1Zr, 8.9 g/cm³), which has excellent high-temperature strength

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[27–30]. The microstructure and precipitates in different regions of EBSM-built Nb521 alloy along the building direction were carefully examined and compared with its ingot. The morphology, composition and formation mechanism of precipitates were systematically analyzed in combination with the thermal history during the EBSM process, providing an experimental basis for the further applications and optimization of EBSM-built refractory alloy in various fields.

2. Materials and methods

Spherical powders prepared by the plasma rotation electrode process (PREP) were used in this experiment. The surface morphologies, particle size distribution and cross-sectional microstructure of Nb521 powders were shown in Fig. 1. Most of as-received powders possessed a spherical shape, as can be seen in Fig. 1(a) and insert image I, with the particle size distribution ranging from 45 to 105 μm (insert image II). The cross-sectional microstructure of the powders with a dendrite-type structure was shown in Fig. 1(b). The chemical composition of as-received powders was given in Table 1. The EBSM of Nb521 was performed on the QBeam Lab system developed by Tsinghua University, China [31]. During the EBSM process, the chamber vacuum pressure was controlled at $\sim 2 \times 10^{-3}$ mBar to minimize oxygen pick-up, and the stainless-steel substrate was preheated by the electron beam up to 1223 K prior to the melting process. The detailed process parameters can be found in Ref. [32]. Pre-solidified layers will be re-scanned and re-melted with the identical melting parameters before the next spreading to avoid spheroidization and unevenness, as shown in Fig. 2(a). Rectangular samples of $20 \times 20 \times 10 \text{ mm}^3$ dimension were fabricated under beam current ranging from 6 mA to 16 mA and scanning speed ranging from 0.1 m/s to 0.7 m/s. The relative density of the EBSM-built Nb521 samples were measured by the Archimedes method. For nearly fully dense samples, three metallographic specimens of $5 \times 1 \times 5.5 \text{ mm}^3$ dimension, hereafter labeled by S1, S2 and S3, were electrical discharge machined from the top, the middle, to the bottom, respectively, parallel to the building direction at equal intervals, as illustrated in Fig. 2(b), and followed standard metallographic preparation process to observe their microstructure. The identical sampling and metallographic preparation methods were also applied to Nb521 ingot. The microstructure was examined by X-ray diffraction (XRD, PANalytical Empyrean) with $\text{CuK}\alpha$ radiation, scanning electron microscope (SEM, Merlin Compact, 20 kV), and transmission electron microscope (TEM; Talors-F200x; 200 kV). Samples for SEM observations were etched in etchant ($\text{HF}:\text{HNO}_3:\text{H}_2\text{O} = 1:1:1$) for 10s. TEM thin foils were prepared by ion milling on a $\phi 3\text{mm} \times 40 \mu\text{m}$ disc to ensure electron transparency. The length and proportion of precipitates were measured by Image Pro Plus (IPP)

software. The three-dimensional (3D) surface morphologies of the samples were observed by confocal laser scanning microscope (CLSM, OLYMPUS).

3. Results and discussion

3.1. EBSM processability of Nb521 alloy

The effect of electron beam current and scanning speed on the surface morphologies of EBSM-built Nb521 samples was shown in Fig. 3(a), and the parameters indicated by the same symbol represented a similar upper surface. Porous surface, flat surface and wavy surface were involved in these samples, corresponding to the insufficient energy input, moderate energy input and excessive energy input during the EBSM process, respectively. An appropriate processing window can be identified after initial processing trials. As can be seen in Fig. 3(a), flat upper surface of EBSM-built Nb521 alloy samples can be obtained under a scanning speed of 0.7 m/s and beam current ranging from 13 mA to 16 mA, and the outer appearance was shown in Fig. 3(b). The sample fabricated at 15 mA had the smoothest surface as the beam current increases from 13 mA to 16 mA. The 3D surface morphologies of these samples in Fig. 3(b) were measured by CLSM, as shown in Fig. 3(c). Cloud maps in Fig. 3(c) indicated the flatness of the sample surface, and a decrease in the contrast of the cloud map suggested an improvement in the flatness. The contrast of the cloud map at 15 mA tended to be identical, indicating no significant valleys or peaks on the sample surface [33]. The relative density of these samples in Fig. 3(b) fluctuated between 96.15% and 99.19%, as shown in Fig. 3(d), and a near-full dense sample was obtained under a beam current of 15 mA.

The upper surface morphology and internal defects of EBSM-built samples were closely related to the flow and solidification of the molten pool. The molten pool was strongly dynamic and its fundamental behavior was governed by capillary forces in combination with wetting [34,35]. The solidification morphology of the molten pool was strongly dependent on the wetting characteristics of the molten pool with the powders. As the insufficient energy applied, the molten pool was small and powders near the molten pool boundary were partially melted. Driven by the surface tension, a solidified connection was formed between the melted portion of the powders and the molten pool [36], and pores between them remained after solidification. Moreover, powders close to the molten pool boundary will be dragged into the melted region, as shown in the insert images in Fig. 3(d) under 13 mA and 14 mA. When the moderate energy input was adopted, the fluctuation of the molten pool was relatively stable and powders were fully melted, ensuring no pores inside the sample and a flat surface after

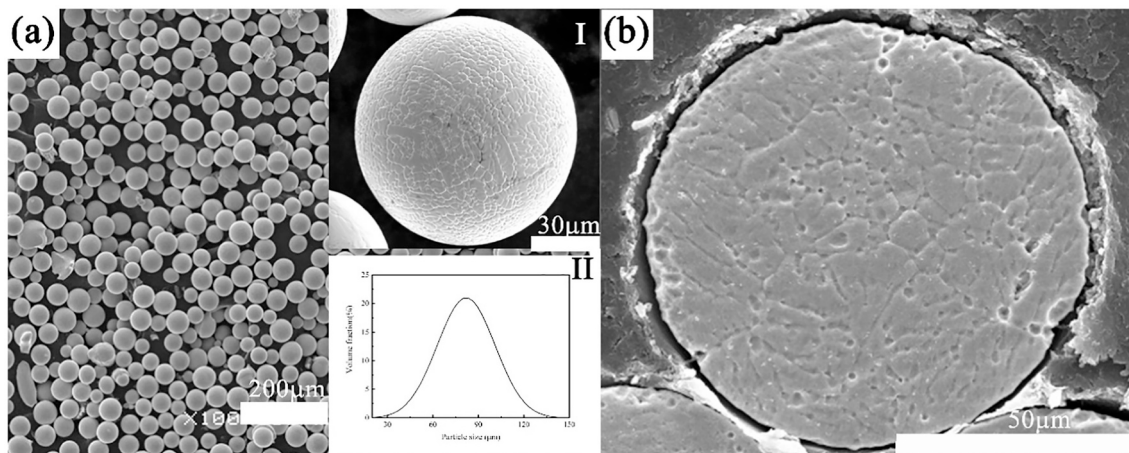


Fig. 1. (a) Surface morphology of Nb521 alloy powders produced by PREP, with insert image I illustrating the enlarged surface morphology and image II highlighting the particle size distribution, and (b) the cross-sectional microstructure of a typical Nb521 alloy powder.

Table 1

Chemical compositions (wt%) of PREPeb Nb521 alloy powders and ingot.

Elements	W	Mo	Zr	Ta	C	O	N	H	Nb
Powder	4.94	1.98	1.65	0.09	0.006	0.015	0.005	0.0006	Bal.
Ingot	5.03	2.10	1.70	0.08	0.0029	0.025	0.006	0.0011	Bal.

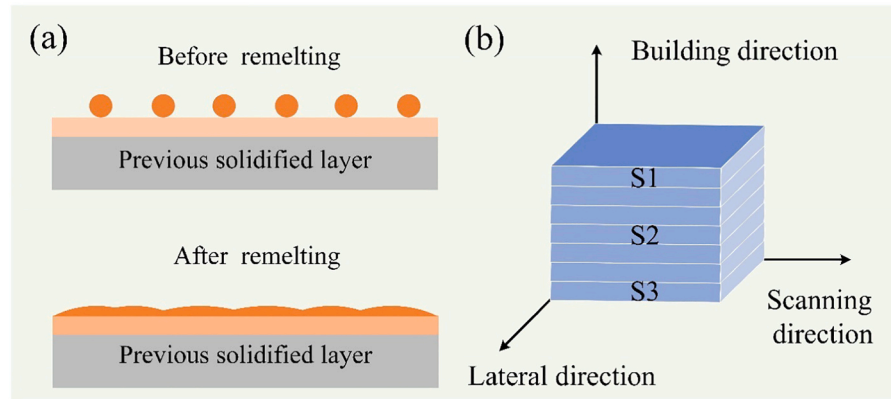


Fig. 2. (a) Schematic diagram of spheroidization of Nb521 powders before and after re-melting, indicated that the re-melting of the previous solidified layer was beneficial to improve surface forming quality, and (b) schematic diagram of sampling positions S1, S2, S3 of the EBSM-built Nb521 alloy for structural observations.

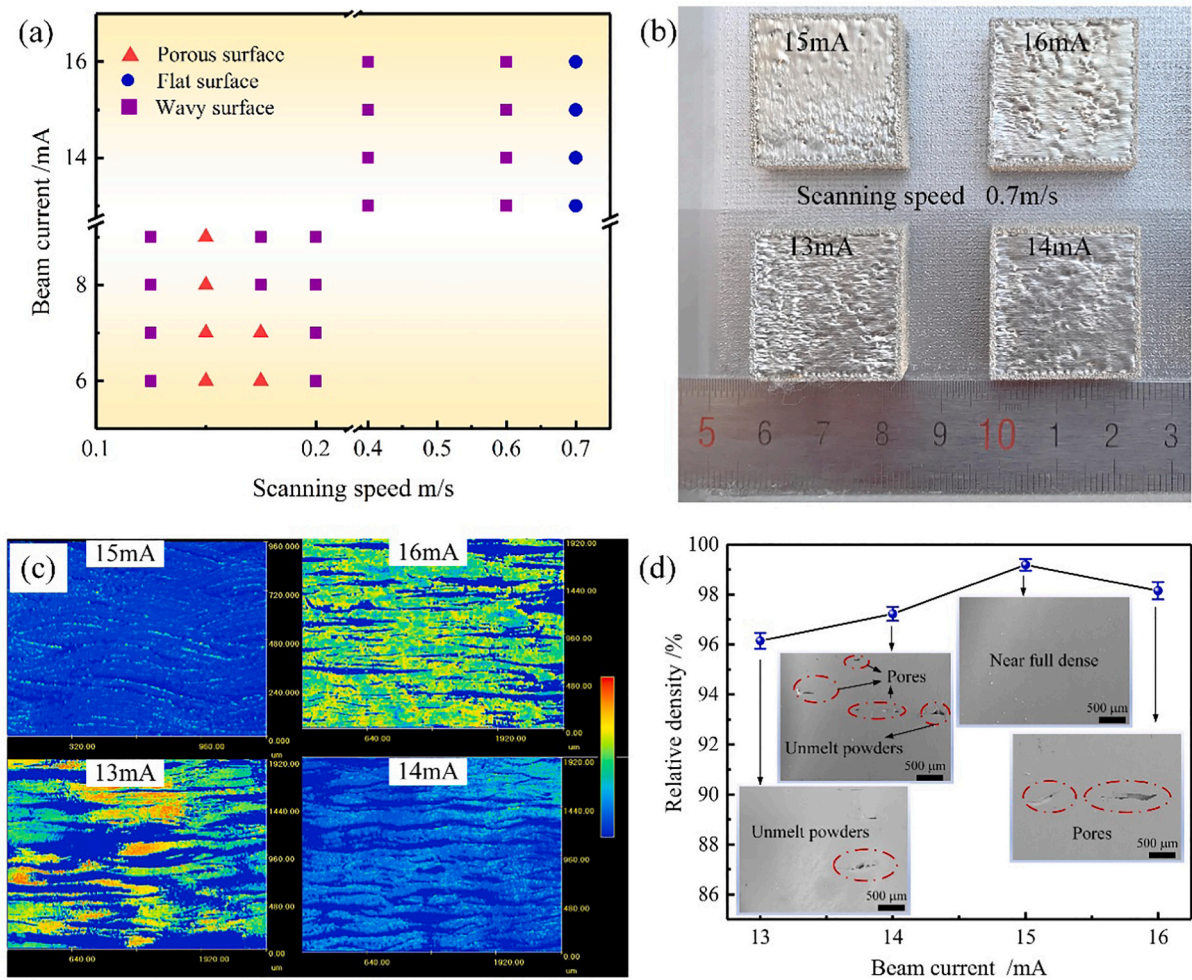


Fig. 3. (a) Process map to obtain moderate process window and (b) the outer appearance of EBSM-built Nb521 samples under a scanning speed of 0.7 m/s and beam current of 13 mA, 14 mA, 15 mA and 16 mA. Influence of beam current on the (c) surface morphology and (d) the relative density of the EBSM-built Nb521 samples.

solidification. The excessive energy input inevitably caused the vaporization and violent fluctuation of the molten pool. The resulting vapor pressure lead to the imbalance of liquid metal surface tension, resulting in the collapse of the depression and pores in the molten pool [37]. Under the condition of excessive energy input, it was easy to encapsulate metal vapor, favoring the formation of cavities [38], as shown in the insert image in Fig. 3(d) under 16 mA.

3.2. Microstructures under different conditions

The vertical cross-section micrographs of EBSM-built Nb521 alloy samples under various electron beam currents were examined in the top, middle and bottom regions, and the results were shown in Fig. 4. In all cases, white precipitates were distributed discretely in rod-like or dot-like morphologies on the Nb solid solution (Nbss) matrix. SEM images of top region were shown in Fig. 4 (a), (d), (g) and (j), where the precipitates were rod-like with high aspect ratio. In the middle region, the elongated precipitates were bent or broken, and a small amount of dot-like precipitates appeared. In the bottom region, the aspect ratio of rod-like precipitates decreased whereas the number of dot-like precipitates increased, and all the precipitates were homogeneously distributed on the Nbss matrix. The content of precipitates gradually increased from the top to the bottom regions for the sample under the same parameters. The morphology and size of precipitates varied significantly between different regions suggesting that there was a significant difference in thermal history between different layers.

Driven by capillary forces, the morphology of precipitates upon elevated temperature exposure will change and eventually lead to spherical dispersoid shape [39]. The evolution in precipitates morphology was schematically depicted in Fig. 5. For rod-like precipitates with a high aspect ratio, the shape was unstable with respect to the development and growth of perturbations. When the system was heated sufficiently to allow atomic migration, thermal grooves were

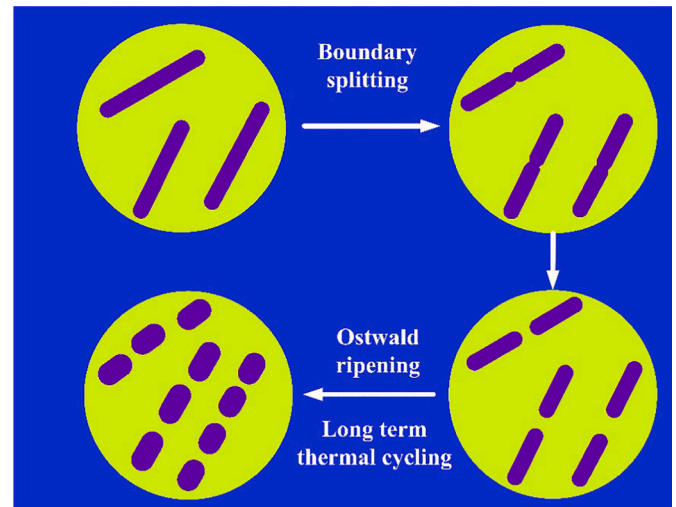


Fig. 5. Schematic diagram illustrating the evolution of precipitates morphology during long term thermal cycling.

formed in order to establish static equilibrium among surface tensions [39,40]. The formed grooves were surrounded by two ridges, and the curvature of these ridges promoted the mass transport away from the groove, further promoting the deepening of the grooves to achieve the re-establishment of tension balance. Precipitates split when the thermal grooves on the boundary were thick enough to penetrate the entire precipitate. During the long-term thermal cycle, fractured precipitates will become larger under the effect of Oswald ripening. Moreover, precipitates can be fully precipitated with prolonging the heat preservation time, resulting in an increase in the precipitate content.

The microstructure characteristics of EBSM-built Nb521 alloy

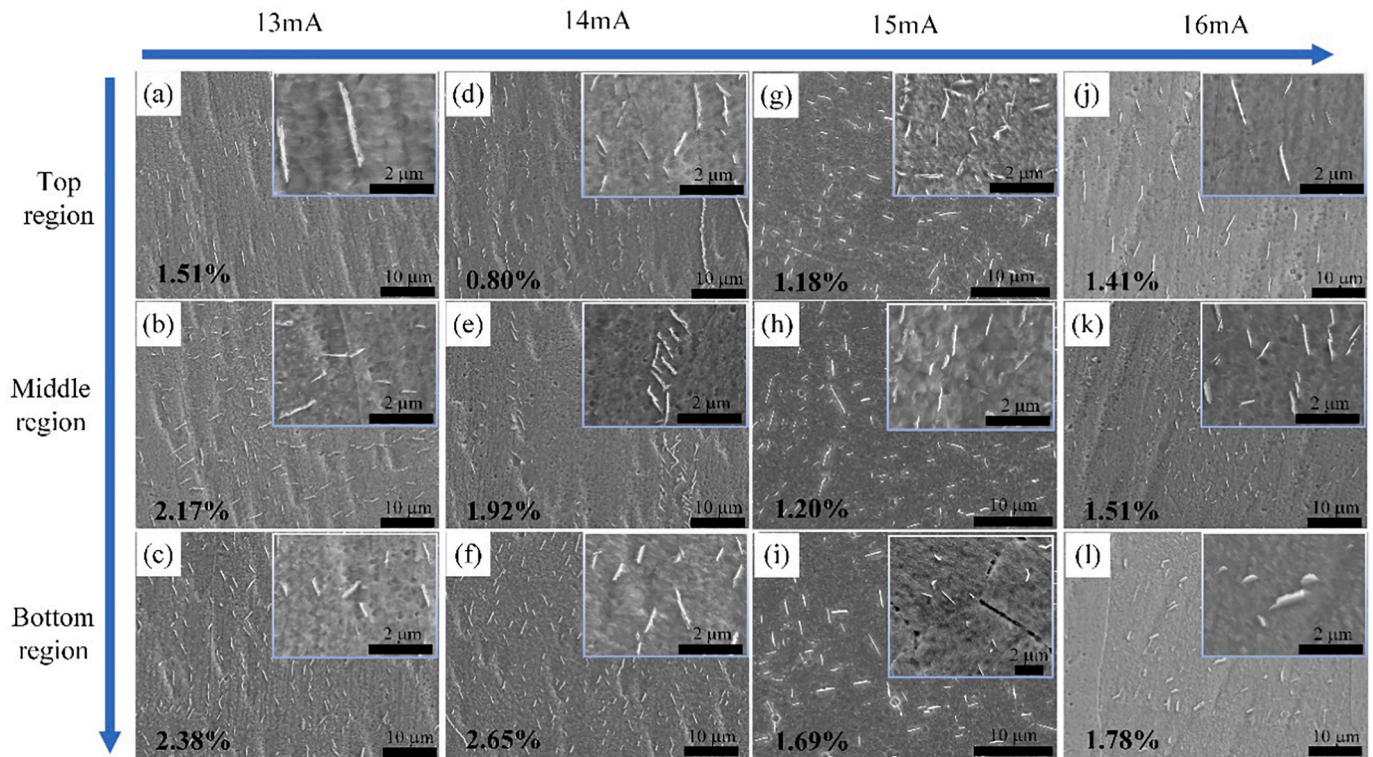


Fig. 4. Typical cross-sectional microstructure images at the top, middle and bottom region of the EBSM-built Nb521 samples with different beam current (a-c) 13 mA, (d-f) 14 mA, (g-i) 15 mA and (j-l) 16 mA. The insert image shows the magnified image of the precipitates, and data in each image represents the proportion of precipitates.

samples under 15 mA current were presented in Fig. 6. Columnar grains and equiaxial grains can be seen in the vertical [see Fig. 6(a), (c)] and transverse cross-section [see Fig. 6(b), (d)], and clear fusion boundaries can be seen in all images. The distance between adjacent molten pool boundaries in vertical cross-section was about 50 μm , as shown in Fig. 6(a), less than the thickness of each powder layer (75 μm), indicating that the penetration depth of the electron beam exceeded several powder layers thickness [41]. The distance between adjacent molten pool boundaries in the transverse cross-section was about 200 μm , equal to the scanning line spacing, as shown in Fig. 6(b). The content of precipitates in larger grains was larger than that in finer grains, whereas there were almost no precipitates in ultrafine grains. Moreover, white precipitates were distributed along the fusion boundaries, making the fusion boundaries more obvious, as shown in Fig. 6(c)–(d). The partial enrichment of precipitates was mainly due to the fact that in the initial stage of molten pool formation, the heat transfer in the molten pool mainly depended on heat conduction. Owing to the insufficient mixing time and low convection, elements in the molten pool were inhomogeneously distributed. With the development of the molten pool, the mass transfer in the molten pool was dominated by strong Marangoni convection. Under the combined action of Marangoni tangential force and thermal buoyancy, reflux occurred in different regions of the molten pool [42]. Since the temperature gradient in the upper region was larger than that in the bottom region of the molten pool, the reflux in the upper region was intenser, and elements tended to accumulate on the

periphery of the molten pool. Elements concentration at the rear edge and the periphery of the molten pool were higher than that at other positions [42,43]. Meanwhile, an unmixed zone was observed at the bottom of the molten pool. In the solidification process of multicomponent alloys system, microsegregation existed in the molten pool due to the banding during molten pool solidification, causing perturbations in the solidification structure and the solute concentration [44,45]. Due to the high cooling rate, the diffusion of elements into the solid phase was incomplete in multi-component alloys. During the layer-by-layer accumulation process, the solutes within the molten pool were redistributed. For the remelting process, the concentration at the bottom of pre-solidified layers remained since the remelting only occurred at the top part of the pre-solidified layer. The heterogeneous distribution of these elements at the bottom of the molten pool led to the enrichment of precipitates along the fusion boundary.

3.3. Evolution of precipitates during EBSM process

3.3.1. XRD phase identification

The XRD patterns of EBSM-built Nb521 samples and Nb521 ingot were shown in Fig. 7. Four samples exhibited almost the same diffraction peak positions, mainly containing Nbss diffraction peaks. In the Nb–W–Mo–Zr system alloy, substitutional solid solution was easily formed in W–Nb and Mo–Nb. No separate diffraction peaks of W and Mo were found, indicating that W and Mo dissolved in the Nbss. The

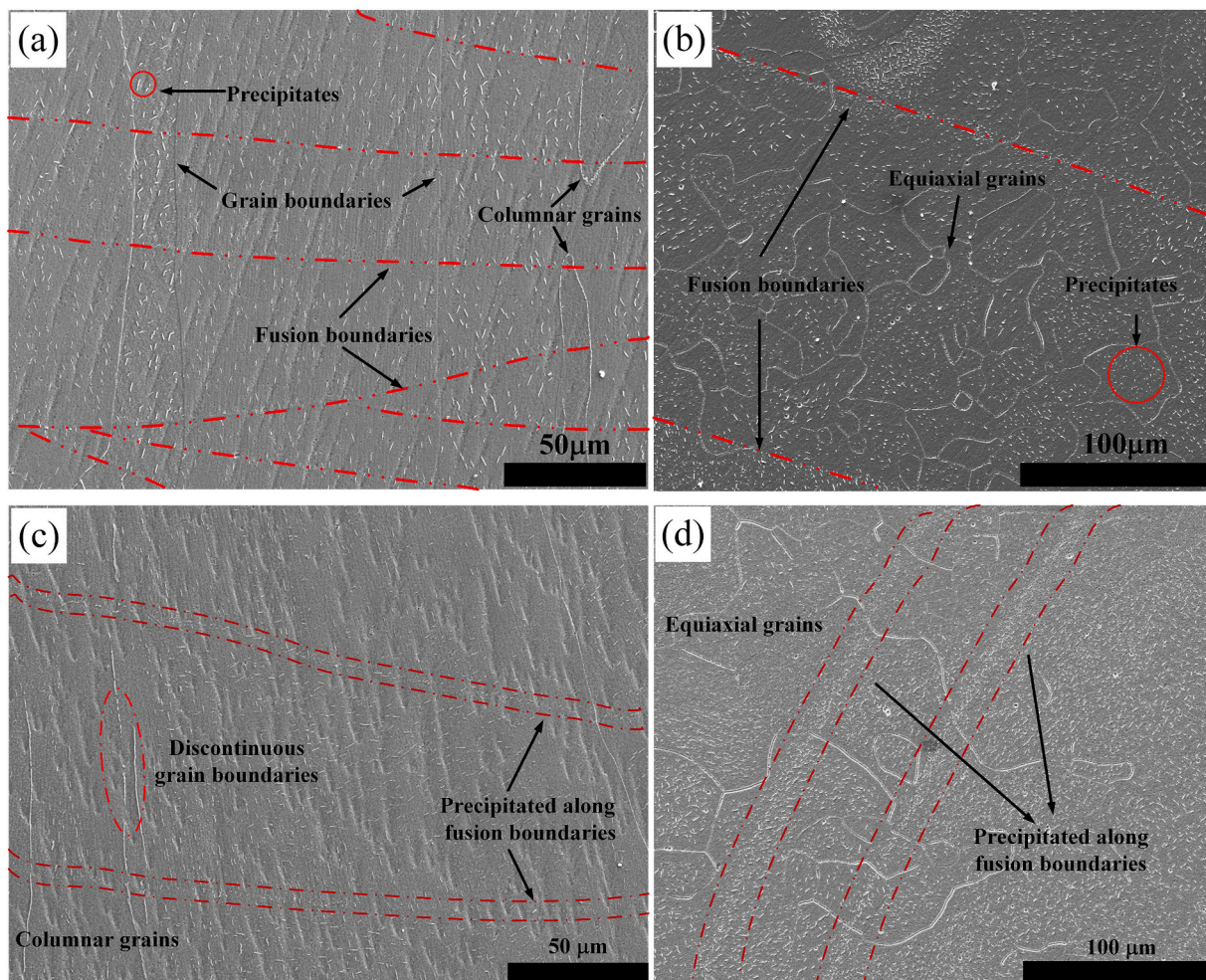


Fig. 6. SEM images of the EBSM-built Nb521 samples under a beam current of 15 mA showing clear fusion boundaries, columnar or equiaxial grains and white precipitates in (a) vertical and (b) transverse cross-section. Precipitates along melt pool boundaries and grain boundaries in (c) vertical and (d) transverse cross-section.

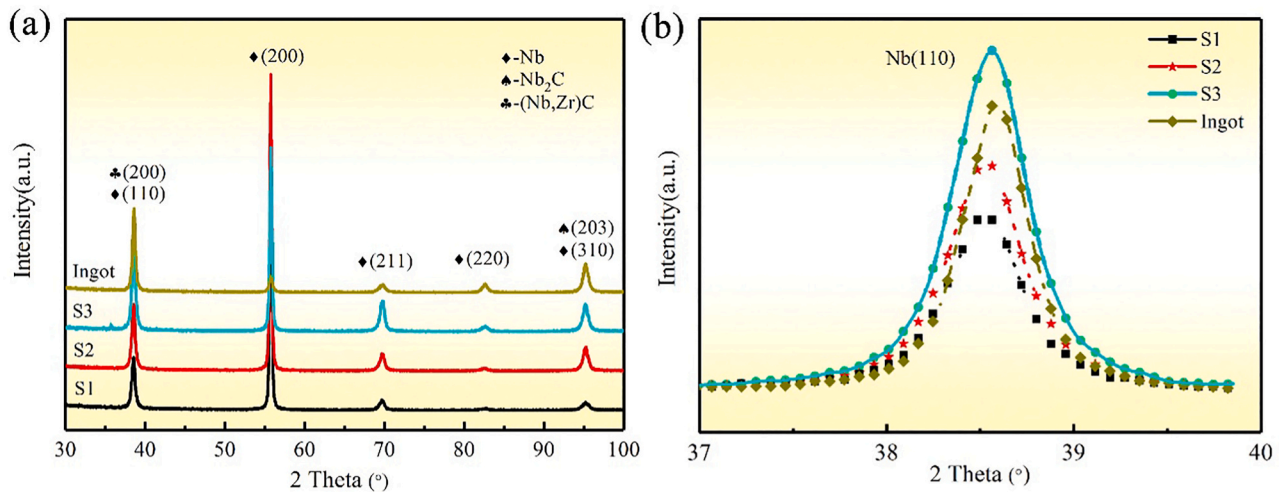


Fig. 7. (a) XRD patterns and their enlarged patterns (b) highlighting Nb(110) peak of EBSM-built samples in different regions and Nb521 ingot samples.

Nb₂C and (Nb, Zr)C precipitates were distributed on (203) and (200) crystal planes, respectively, as shown in Fig. 7(a). EBSM-built Nb521 alloy samples had a certain offset in (110) peak positions, compared with the Nb521 ingot (Fig. 7(b)). EBSM-built Nb521 samples exhibited the highest intensity on the (200) crystal plane, indicating that grains grow preferentially along this direction. The diffraction peak intensity from S1 to S3 samples gradually increased. Since the atomic radius of W and Mo was slightly smaller than that of Nb, the lattice constant of Nb atom decreased as W and Mo dissolved into Nb atom, suggesting a diffraction peak shift to higher angle. Contrarily, the lattice constant of Nb increased as Zr atoms dissolved into Nb atoms. In fact, no obvious diffraction peak offset can be found in Fig. 7, except for (110) crystal plane, which can be attributed to the combined effect of the above two phenomena.

3.3.2. Microstructure examination

Typical SEM images of EBSM-built Nb521 samples at different positions parallel to building direction, together with that of Nb521 ingot, were shown in Fig. 8, with the black and the white area corresponding to the Nbss and precipitates, respectively. A large number of corrosion pits, which were mainly carbides, were distributed intra-granular and along grain boundaries, and the sizes of corrosion pits at grain boundaries were larger than that of intra-granular. This can be attributed to the greater atomic diffusion at the grain boundary than that intra-granular under identical conditions. In EBSM-built Nb521 samples, dot-like or rod-like precipitates with several micrometers in length were discretely distributed, as shown in Fig. 8(a)-(c), whereas long needle-like precipitates with large aspect ratio existed in ingot samples (Fig. 8(d)). The average size of precipitates of S1, S2, S3 and ingot were measured to be 1.14 μm , 0.92 μm , 0.78 μm and 117.40 μm (Fig. 8(e)), respectively.

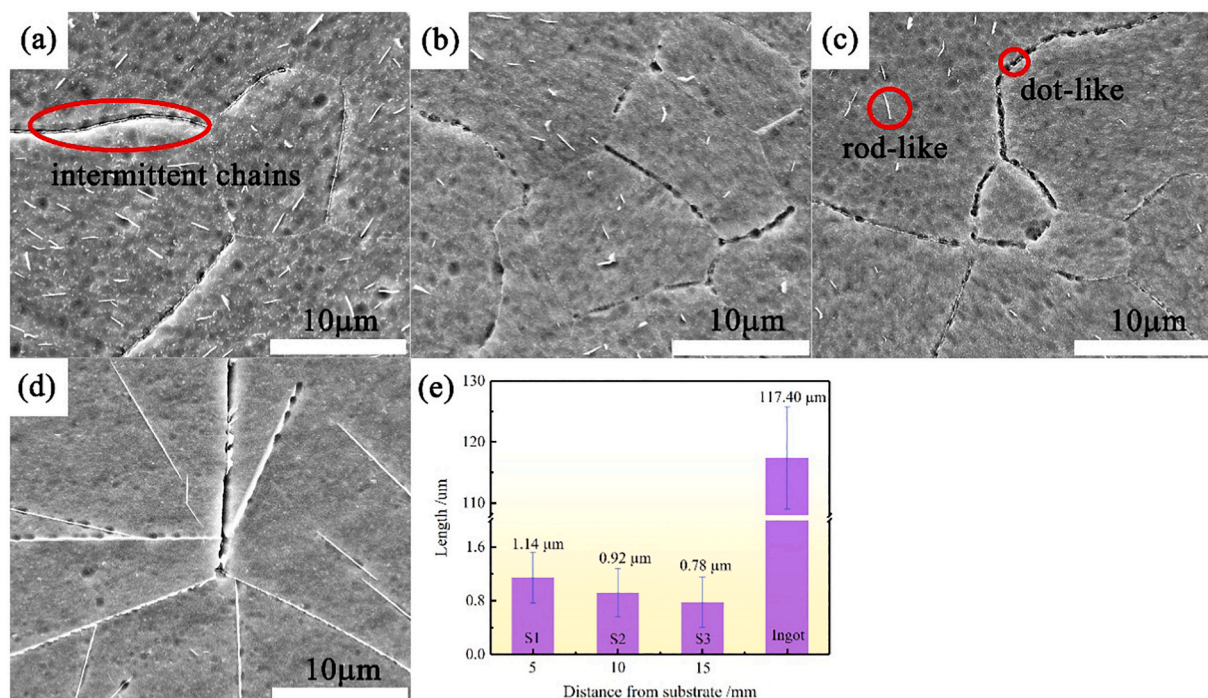


Fig. 8. SEM images of EBSM-built Nb521 sample (a) S1, (b) S2, (c) S3, (d) ingot samples, and (e) the average length of precipitates in EBSM-built Nb521 and ingot samples.

Compared with ingots, ultrafine precipitates and grains were formed under the extremely high cooling rate in the EBSM process. The difference in the grain size of ingot and EBSM-built Nb521 alloys can be ascribed to the difference in diffusion rate and nucleation energy during these two fabrication processes [46]. With the decrease in the thermal cycle numbers and insulation time from S3 to S1, the content and size of precipitates gradually increased and precipitated in an intermittent chain along the grain boundary, as shown in Fig. 8(a). The difference in morphology and volume fraction of precipitates will result in the microstructure inhomogeneity of the EBSM-built Nb521 alloy sample [47].

To further determine the phase composition, high-angle annular dark field (HAADF) observations were carried on dot-like and rod-like precipitates in the S2 sample, as shown in Fig. 9. The quantitative

analysis of the chemical compositions of precipitates along the grain boundary and intra-granular were performed, and the corresponding results were listed in Table 2. As shown in Fig. 9(a)-(c), dot-like precipitates along the grain boundary were rich in Zr, and the detected

Table 2

Chemical compositions (at.%) of the precipitates of EBSM-built Nb521 alloy.

Elements	Nb	W	Mo	Zr	C
Dot-like phase (Zr enrichment)	43.99	1.54	1.24	37.91	17.10
Dot-like phase (C enrichment)	55.64	4.87	1.52	0.82	34.06
Rod-like phase	54.88	4.97	1.31	0.89	37.95

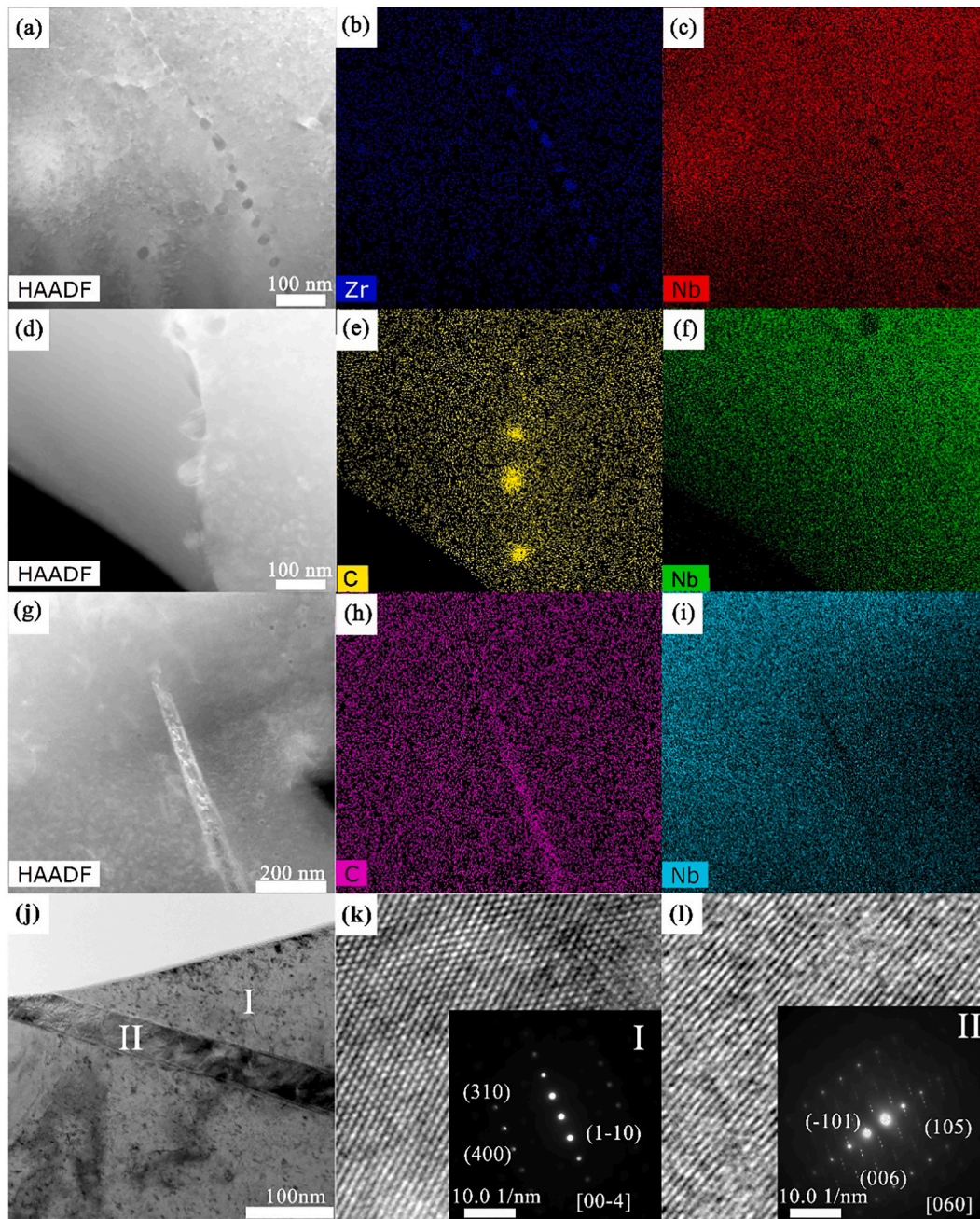


Fig. 9. Typical microstructure and element distribution maps of S2 sample: TEM bright field image of (a) Zr-rich, (d) C-rich dot-like precipitates at the grain boundary and (g)(j) C-rich rod-like precipitates in the grains, and element distribution map of (b) Zr, (e)(h) C, (c)(f)(i) Nb, and HRTEM and SAED (inset) of (k) region I and (l) region II in (j).

atomic ratio of Nb, Zr and C was close to 1:1:1, determined as (Nb, Zr)C. Meanwhile, the identified elements in some dot-like precipitates along the grain boundary consisted of C and Nb elements, as shown in Fig. 9 (d)-(f), and the Nb, C atomic ratio was about 2:1. Combined with XRD results, these dot-like precipitates were determined as Nb₂C phases. The intra-granular rod-like precipitates were mainly rich in C elements, as shown in Fig. 9(g)-(i), and the Nb, C atomic ratio was about 2:1, which further verified the formation of Nb₂C phases.

Based on the binary phase diagram and thermodynamic theory [48], the Gibbs free energy for the formation of compounds between Nb, W, Mo and Zr were higher than those of carbides, such as (Nb, Zr)C or Nb₂C. The dependence of the standard Gibbs free energy (ΔG^0 , kJ/mol) of these reactions on the temperature (T , K) were expressed as follows [49]:

$$\text{Nb(s)} + \text{C(s)} = \text{NbC(s)}, \Delta G^0 = -136900 + 2.43 T \quad (T < 1500 \text{ K}) \quad (1)$$

$$2\text{Nb(s)} + \text{C(s)} = \text{Nb}_2\text{C(s)}, \Delta G^0 = -193700 + 11.7 T \quad (T < 1500 \text{ K}) \quad (2)$$

$$\text{W(s)} + \text{C(s)} = \text{WC(s)}, \Delta G^0 = -42,260 + 4.98 T \quad (1302 < T < 1400 \text{ K}) \quad (3)$$

$$2\text{W(s)} + \text{C(s)} = \text{W}_2\text{C(s)}, \Delta G^0 = -30540 - 2.34 T \quad (900 < T < 1302 \text{ K}) \quad (4)$$

$$\text{Mo(s)} + \text{C(s)} = \text{MoC(s)}, \Delta G^0 = -75,300 - 5.44 T \quad (T < 700 \text{ K}) \quad (5)$$

$$2\text{Mo(s)} + \text{C(s)} = \text{Mo}_2\text{C(s)}, \Delta G^0 = -45600 - 4.18 T \quad (T \geq 700 \text{ K}) \quad (6)$$

$$\text{Zr(s)} + \text{C(s)} = \text{ZrC(s)}, \Delta G^0 = -196650 + 9.2 T \quad (25 < T < 1850 \text{ K}) \quad (7)$$

Therefore, the precipitates were mainly dominated by carbides. Carbides in Nb—W alloy were generally precipitated during solidification or heat treatment, mainly including rod-like, dot-like or needle-like first-order carbides NbC, and ZrC [50], and second-order carbides Nb₂C. For the EBSM process, the cooling rate of the molten pool can reach up to 10^3 – 10^6 K/s [51], demonstrating non-equilibrium solidification features. During the rapid solidification of the molten pool, the primary NbC and ZrC carbides were formed firstly. As Nb atoms dissolved massively in ZrC, (Nb, Zr)C was formed. With decreasing the

temperature, some carbon atoms in the metastable carbide phase precipitated and formed stable Nb₂C carbides with the surrounding Nb atoms. The dynamic thermal balance in pre-solidified layers and thermal insulation in the high vacuum environment during the layer-by-layer printing process also promoted the formation of Nb₂C carbides.

TEM observations have been performed on rod-like precipitates at S2. Fig. 9(j) shows the bright-field TEM image of the S2 sample. The width of the precipitates was measured to be ~50 nm. Fig. 9(k) and Fig. 9 (l) shows the high resolution TEM images of the matrix and the rod-like precipitate marked by I and II in Fig. 9(j), respectively, with inset showing their corresponding selected area electron diffraction (SAED) patterns. As seen, there was a great difference in the atomic arrangement between the matrix and the rod-like precipitates. According to the SAED patterns, the matrix and precipitate was identified to be Nbss [Fig. 9(k) inset], and Nb₂C [46] [Fig. 9(l) inset], respectively.

3.3.3. Evolution of precipitates during EBSM process

Fig. 10 illustrated the thermal history and precipitates evolution during the EBSM process. Fig. 10(a) showed the typical thermal cycles of S3 sample, with Δt representing the duration under thermal balance, where the difference lies in the variation of Δt . Here, Δt_1 , Δt_2 and Δt_3 represented the thermal balance holding time at S1, S2 and S3 in Fig. 8 (a)-(c), respectively, in the order of $\Delta t_1 < \Delta t_2 < \Delta t_3$. Thermal cycles with decreasing amplitudes in Fig. 10(a) represented the rapid heating and cooling of the EBSM-built sample as subsequent layers built. During long-term thermal accumulation, the internal temperature will fluctuate within a small range and remain for a period of time, in which EBSM-built samples were subjected to annealing-like heat treatment.

Fig. 10(b)-(d) showed the evolution of the precipitates during the EBSM process. At high temperature, interstitial carbon atoms diffused into Nb atoms to form carbide precipitates such as NbC, Nb₂C and NbC_{1-x}, and stabilize to 3600 °C according to the Nb—C phase diagram [48,52]. Due to the low carbon content in the EBSM-built Nb521 sample, Nb₂C was more likely to precipitate, and the difference in the growth rate of carbides under different temperature environments can favor the formation of gradient microstructures [46]. Analogously, the inter-diffusion between W and Nb will occur as the sample cools down from

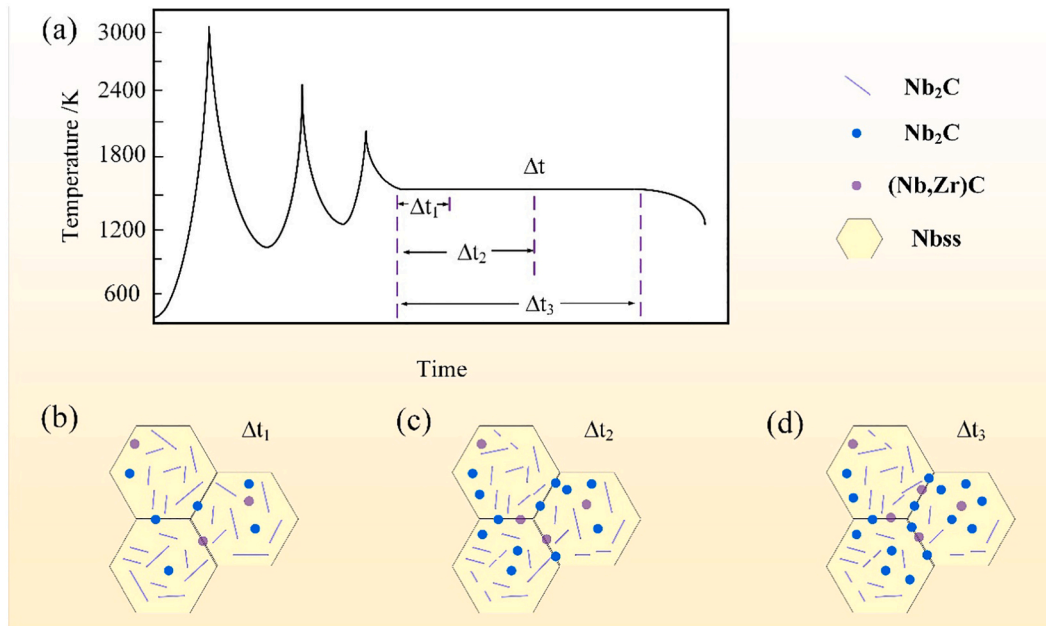


Fig. 10. Schematic diagram of thermal history and precipitates evolution during EBSM process: (a) thermal history of EBSM-built Nb521 alloy, (b) precipitates of EBSM-built Nb521 alloy after short-term thermal cycling Δt_1 , with long and dot-like phases distributed within the grain and along the grain boundaries, respectively, (c) precipitates of EBSM-built Nb521 alloy after entering the thermal balance stage and holding for a period of time Δt_2 , and (d) precipitates of EBSM-built Nb521 alloy after a long period of insulation Δt_3 under thermal balance.

high temperature [53]. It can be seen from the Zr–C phase diagram that, when C and Zr were added to the Nb–W refractory alloy, ZrC or (Zr, Nb)C phases can be precipitated from the matrix. As shown in the Nb–W–Mo ternary phase diagram [54], both Mo and W can form infinite solid solutions, but the Gibbs free energy of the reaction of C–Nb and Zr–Nb was lower than those of other reactions. Therefore, compounds formed by the reaction of Zr–Mo, Zr–W, C–W and C–Mo were difficult to be observed in the EBSM-built sample. Consequently, all the precipitates involved in EBSM-built Nb521 samples contained two carbide phases: Nb₂C, and (Nb, Zr)C [see Fig. 7(a)].

As the sample entered the dynamic thermal balance initial stage, *i.e.* Δt_1 , as shown in Fig. 10(b), dot-like or rod-like precipitates were distributed inside the grains or along the grain boundaries. Carbides, such as Nb₂C and (Nb, Zr)C, were precipitated in an intermittent chain along the grain boundary, as shown in Fig. 8(a). When the holding time Δt lasted until Δt_2 (Fig. 10(c)), part of the elongated Nb₂C was broken up, causing a decrease in the aspect ratio of broken precipitates and thus favoring the formation of finer precipitates, as shown in Fig. 8(b). As the AM process proceeded, the annealing effect of pre-solidified layers caused by prolonging the holding time became more and more obvious, and the precipitates became spherical and uniform due to Ostwald ripening, as shown in Fig. 10(d) and Fig. 8(c), which was beneficial to the improvement of strength and creep performance. Even though (Nb, Zr)C was unstable under high temperature and easily decomposed into Nb₂C, some residual dot-like (Nb, Zr)C can be still seen in the as-built Nb521 sample, as shown in Fig. 9(a), which may be attributed to the incomplete decomposition resulting from the extremely high cooling rate.

4. Conclusions

The refractory Nb521 alloy was fabricated by EBSM. The evolution mechanism of precipitates in combination with thermal history inherent in the EBSM process was elaborated. The following conclusions can be drawn:

- (1) The feasibility of refractory Nb521 alloy fabricated by EBSM was verified, and EBSM-built Nb521 with a high relative density of >99.19% can be obtained at an electron beam current of 15 mA and a scanning speed of 0.7 m/s.
- (2) Long rod-like and dot-like precipitates were distributed intra-granular and along the grain boundaries. Ultrafine dot-like precipitates that enrich in Zr and C elements were determined as (Nb, Zr)C and Nb₂C, and rod-like precipitates enriched in C elements were determined to be Nb₂C.
- (3) As the thermal cycle numbers increased and the holding time prolonged, the morphology of precipitates at the grain boundary changed from the intermittent chain to dot-like, whereas part of the elongated precipitates inside grains was broken up and became spherical and uniform.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical limitations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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